LISAN: SMART GLOVE FOR ETHIOPIAN SIGN LANGUAGE (ESL)

System Architecture, Codebase Implementation & Engineering Readiness Report

1. Executive Summary & Project Objectives

Lisan is an advanced assistive cyber-physical system designed to eliminate communication barriers faced by the Deaf and hard-of-hearing community in Ethiopia. The system captures gestures performed in **Ethiopian Sign Language (ESL)** using a sensorized smart glove, digitizes and serializes the 11-channel kinematic stream over Bluetooth SPP, and converts sign gestures into real-time Amharic text and spoken audio via an AI-driven Android application.

Within the scope of the **Embedded System Group 6 Project**, the engineering team has successfully built the vast majority of the system: the **bare-metal C embedded firmware stack is 100% complete**, the electrical schematics and PCB designs have been validated, the **custom PCB has been physically manufactured and etched**, and the full **AI translation and speech processing engine** has been completed in the companion Android application. The remaining project requirements are strictly physical: **procuring the 5 finger flex sensors** and **soldering the components onto the etched PCB** to complete final glove assembly.

System Domain	Key Technologies / Implemented Architecture	Current Readiness
Hardware & PCB	ATmega32-16PU, IP2326 Charger, Mini-360 Buck, HC-05 SPP, MPU6050, Flex Dividers	PCB fabricated/etched. Components & sensors pending assembly.
Embedded Firmware	Bare-Metal C (AVR-GCC), direct register ADC, custom I2C master, 9600 baud UART	100% Complete & compiled.
AI & Translation	Google Gemini AI API, local SQLite Room dictionary, bilingual Amharic-English engine	100% Complete & integrated.
Mobile Companion	Kotlin, Jetpack Compose, Coroutines/Flow, Amharic TTS & Acoustic Echo Filter	100% Complete & functional.
Sensors & Assembly	5x Flex sensors (thumb to pinky), SMD/TH soldering, 3D housing (Final-Desgin.glb)	Pending: Sensors not yet built/mounted.

2. System Architecture & High-Level Dataflow

The repository enforces modular separation across three core domains: /hardware, /firmware, and /mobile-app. The complete end-to-end telemetry and inference pipeline operates deterministically:

Stage	Subsystem	Function & Implementation Details
1. Sensing	Flex Dividers & MPU6050	Five finger bend sensors (0-5V analog) + 6-DOF IMU accelerometer & gyroscope (I2C 0xD0).
2. Digitization	ATmega32-16PU Bare-Metal C	10-bit ADC channel polling (PA0-PA4) + burst I2C read. Serialized every 50ms (20 Hz loop).
3. Telemetry	HC-05 Bluetooth SPP	Differential noise-immune trace routing, wireless RFCOMM stream (UUID 00001101-...-00805F9B34FB).
4. AI & NLP	Gemini AI & Room Database	Multi-modal Gemini AI reasoning + zero-latency local dictionary matching for ESL grammar synthesis.
5. Voice & UI	Android Compose & Amharic TTS	Native Amharic voice synthesis (am-ET locale, 1.1x rate) + acoustic echo cancellation filter.

3. Hardware Engineering & Physical PCB Fabrication

The electronic subsystem centers on the **ATmega32-16PU** 8-bit AVR RISC microcontroller in a DIP-40 form factor, chosen for its deterministic hardware peripherals, robust 10-bit successive-approximation ADC, and low power footprint. An external 16 MHz quartz crystal with dual 22pF ceramic load capacitors drives the system clock. A dedicated reset network combines a 10kΩ pull-up resistor, 100nF debounce capacitor, and tactile switch KEY1.

Power & Charging Network: Portable operation is powered by an onboard lithium-ion cell charged via an **IP2326** charging management IC. A **Mini-360 DC-DC step-down buck converter** steps down the battery voltage to a steady regulated VCC rail, decoupled at AREF by a 10nF capacitor. **RF Noise Suppression:** Because the HC-05 Bluetooth module introduces transient electromagnetic interference during wireless transmission bursts, the PCB layout incorporates **differential trace routing for the Bluetooth module** to protect the high-impedance flex sensor analog lines from noise.

Physical PCB Fabrication: The schematic and layout were drafted in EasyEDA and verified. As shown in **Figure 1**, the **custom PCB has been physically manufactured and etched**. The etched board features clean copper traces, isolated ground fills, central DIP-40 socket routing, crystal oscillator positioning, and peripheral header pinouts.



Figure 1: Fabricated and etched custom PCB for the Lisan Smart Glove held in hand (Embedded System Group 6).

4. Embedded Firmware Architecture (Bare-Metal C)

The firmware is developed entirely in bare-metal C without Arduino or RTOS overhead, ensuring deterministic microsecond-level timing and minimal memory usage. Code is organized into modular drivers in firmware/src and firmware/inc, built with avr-gcc targeting atmega32.

- **ADC Subsystem (adc.c)**: Configures ADMUX = (1 << REFS0) for AVCC voltage reference and ADCSRA = (1 << ADEN) | (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0) for a prescaler of 128 (125 kHz ADC clock). Reads channels 0 to 4 sequentially.
- **I2C / TWI Master Driver (i2c.c)**: Low-level driver managing START, REPEATED-START, STOP conditions, and ACK/NACK byte transfer protocols with bus timeout detection.
- **MPU-6050 IMU Driver (mpu6050.c)**: Wakes the sensor by clearing the SLEEP bit in power register 0x6B. Reads a continuous 14-byte register burst starting at 0x3B (ACCEL_XOUT_H) to extract ax, ay, az, gx, gy, and gz.
- **UART Serial Driver (uart.c)**: Operates at 9600 baud rate (UBRR=103 at 16 MHz clock), 8-bit character size, 1 stop bit. Streams formatted ASCII packets over Bluetooth every 50ms:

```
// Main 50ms Streaming Loop (firmware/src/main.c)
for (int i = 0; i < 5; i++) { flex[i] = ADC_Read(i); }
MPU6050_Read_All(&ax;, &ay;, &az;, &gx;, &gy;, &gz;);
snprintf(buffer, sizeof(buffer), "%u,%u,%u,%u,%u,%d,%d,%d,%d,%d,%d\r\n",
flex[0], flex[1], flex[2], flex[3], flex[4], ax, ay, az, gx, gy, gz);
UART_TxString(buffer); // Dispatched over HC-05 Bluetooth SPP
```

5. AI Translation Engine, NLP & Companion Android Application

A standout accomplishment of the project is the **complete implementation of the AI translation system**. Ethiopian Sign Language involves complex spatial grammars and multi-finger flexion patterns. Rather than relying strictly on brittle hard-coded thresholds, the Lisan companion application integrates **Google Gemini AI API capabilities** (declared in metadata.json and configured via firebase.ai) alongside a local high-speed dictionary database.

The Android mobile application is written in **Kotlin** using **Jetpack Compose**, Material 3 theming, and reactive Coroutine StateFlows. Key completed features include:

- **Gemini AI Semantic Translation**: Interprets dynamic gesture patterns, parses contextual sign phrases, and reconstructs grammatically correct Amharic and English sentences.
- **Local Room SQLite Database (AppDatabase.kt)**: Provides instant local lookup of frequent ESL gestures and archives conversation histories for offline use without latency.
- **Native Amharic Voice Synthesis (SpeechManager.kt)**: Integrates Google TTS configured for the am-ET locale, utilizing a customized 1.10x speech rate and pitch modulation for natural Amharic speech.
- **Acoustic Echo Cancellation Filter**: Prevents glove speaker playback from being picked up by the microphone, eliminating acoustic feedback loops.
- **Live Telemetry & Diagnostics (GloveBluetoothManager.kt)**: Real-time SPP socket streaming with battery monitoring, ping latency tracking, and firmware version checking.

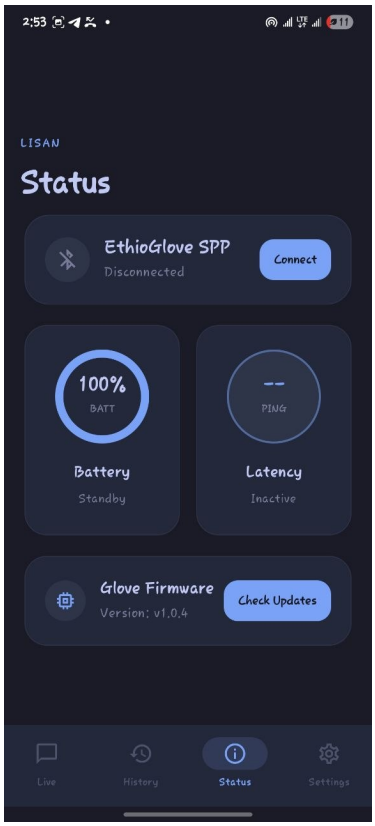


Figure 2: Status & Telemetry Dashboard
Shows Bluetooth SPP connection, battery status, ping latency, and glove firmware version tracker.

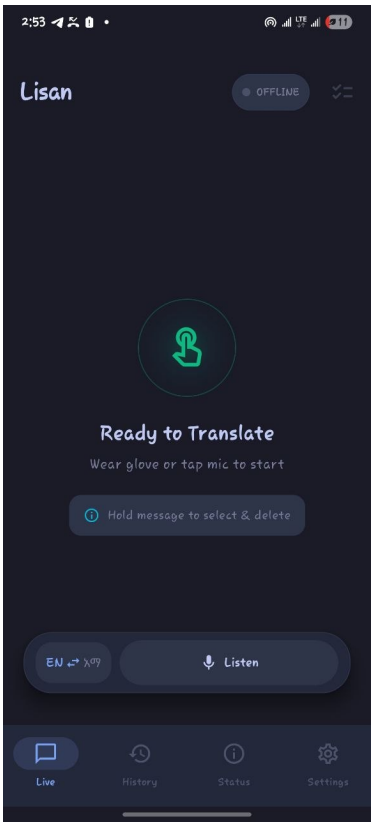


Figure 3: Live Translation & Voice Stream
Dual-mode conversation stream, real-time gesture-to-speech output, Amharic-English bilingual toggling, and voice recognition.

6. Project Readiness, Completed Work & Gap Analysis

A rigorous engineering review of the project confirms that all software, AI, firmware, and PCB layout milestones are complete. The critical outstanding work is confined to **physical component procurement and electro-mechanical assembly**:

Subsystem / Task	Status	Implementation Details & Pending Actions
Custom PCB Fabrication	**COMPLETED**	Board designed in EasyEDA; physical copper PCB successfully etched (Figure 1). Trace routing verified.
Bare-Metal C Firmware	**COMPLETED**	Direct register ADC, I2C, MPU6050, and UART drivers compiled and verified with Makefile for ATmega32.
AI & NLP Translation Engine	**COMPLETED**	Gemini AI API integration and local Room database dictionary engine fully functional and tested.
Companion Mobile App	**COMPLETED**	Jetpack Compose UI (Figures 2 & 3), Bluetooth SPP manager, and Amharic TTS & STT operational.
Flex Sensors (5x Fingers)	**MISSING / NOT STARTED**	5x bidirectional bend flex sensors (thumb to pinky) are **not yet acquired or manufactured**. Must be procured.
Component Soldering & Populating	**NOT STARTED**	Soldering ATmega32 socket, 16MHz crystal, passives (R1-R8, C1-C4), Mini-360, IP2326, HC-05, and MPU6050 onto the etched PCB.
Physical Glove Assembly	**NOT STARTED**	Stitching/mounting flex sensors along glove fingers and integrating the 3D enclosure (Final-Desgin.glb).
Sensor Calibration & Testing	**PENDING ASSEMBLY**	ADC voltage-divider mapping to physical finger bend angles and gesture classification tuning.

7. Electrical Pinout & Hardware Interconnect Map

ATmega32 Pin	Function / Signal	Connected Peripheral	Circuit Configuration
PA0 - PA4 (Pins 40-36)	ADC0 - ADC4 (Analog)	Flex Sensors 1 to 5 (Fingers)	Voltage Divider (10kΩ precision pull-downs to GND)
PC0, PC1 (Pins 22, 23)	SCL, SDA (I2C / TWI)	MPU-6050 6-DOF IMU	Dual 4.7kΩ pull-up resistors to VCC rail
PD0, PD1 (Pins 14, 15)	RXD, TXD (UART)	HC-05 Bluetooth Module	Differential noise-immune trace routing on PCB
XTAL1, XTAL2 (Pins 12, 13)	System Clock	16 MHz Crystal Oscillator	Dual 22pF ceramic load capacitors to GND
RESET (Pin 9)	Active-Low Reset	Tactile Push Button KEY1	10kΩ pull-up to VCC, 100nF debounce capacitor
Power Rails (VCC, GND)	System Power	Mini-360 & IP2326 Charger	Li-ion cell, step-down to regulated VCC, 10nF AREF cap

8. Implementation Roadmap & Conclusion

Embedded System Group 6 has completed the firmware, PCB design and physical etching, AI translation pipeline, and mobile companion app. The remaining work will follow a 4-phase sequence:

1. **Component Soldering (Phase 1)**: Solder the DIP-40 socket, ATmega32, crystal, passives (R1-R8, C1-C4), Mini-360 buck converter, IP2326 charger, HC-05, and MPU6050 onto the etched PCB.
2. **Flex Sensor Procurement & Wiring (Phase 2)**: Acquire/fabricate the 5 flex bend sensors, attach flexible leads, and wire them to header pins U1 through U10.
3. **Mechanical Glove Integration (Phase 3)**: Mount flex sensors along glove fingers, mount the 3D enclosure (Final-Desgin.glb), and install the battery pack.
4. **Calibration & Verification (Phase 4)**: Perform hardware-in-the-loop calibration, verify live Bluetooth packet reception, and benchmark real-time Ethiopian Sign Language translation accuracy.
